

POWERMATIC 1200 RETROFIT — SHOP PACK

The at-the-machine working set · supersedes DWG 01-04 large-format SVGs · pairs with the Rev I ladder set (DWG 05)

Contents

- Sheet 2 · DWG 01 — Power one-line (Rev I: JKS-35 branch fuses in US6J1 blocks; fused Contactor-B tap; fuse map; DBR high; stirring fan)
- Sheet 3 · DWG 02 — Safety relay loop (Rev I: SCM/DCM common caution added; topology unchanged from Rev H)
- Sheet 4 · VFD parameter card — GS23-22P0 (Rev I: DI mode corrected to NPN internal-power; wire 77 deleted)
- Sheets 5-6 · Commissioning & validation checklist (Rev I: fuse map, DI/SCM checks, fan, jog-mode validation deltas)

The working set (implementation documents)

- 1 · BOM (sourcing authority — part numbers in this pack defer to it)
 - 2 · Wire Schedule Rev I (build authority, ferrule = wire no.)
 - 3 · Ladder set Rev I, 6 sheets (programming authority)
 - 4 · This Shop Pack (at-the-machine reference)
- Retired: DWG 03 (subsumed by wire schedule) · DWG 05 SVG (superseded by ladder set) · DWG 04 (archive after panel layout).
The Documentation.md remains the design record and rationale archive; it is not a build document.

Rev I deltas carried in this pack

- GS20 digital inputs: NPN (sink) mode with internal power — factory-default switch. Only common wire is CLICK 0V → DCM (row 81). Wire 77 (+24 → DI common) is DELETED. Never land +24 V on DCM: DCM and the STO common (SCM) share potential inside the drive.
- Contactor B coil tap fused at origin: both 240 V legs through KB-FU (2× GMC-3), rows 22/22A/24/24A.
- Branch fuses: 3× JKS-35 (Class J, fast-acting, 600 V, 200 kA IR) in 3× US6J1 single-pole finger-safe blocks (per BOM).
- DB resistor mounted HIGH in the enclosure; internal 24 V stirring fan added (rows 97-98) — sealed, no penetration, NEMA 12 intact.
- Jog is a MODE (hold-FWD ≥5 s arms; entry hold drives nothing; drum OFF blocks motion, not arming) — see ladder sheets.

Fuse map (single source of truth)

Position	Protects	Block	Element	Rating
Branch (3×)	VFD input branch, panel SCCR	US6J1 ×3 (Class J, finger-safe)	JKS-35	Class J · 35 A · 200 kA IR
PSU-FU (2 poles)	16 AWG PSU-primary tap, 208 V L-L	DN-F10MN ×2	GMC-3 ×2	3 A · time-delay · 250 VAC
KB-FU (2 poles)	16 AWG Contactor-B coil tap, 208 V L-L	DN-F10MN ×2	GMC-3 ×2	3 A · time-delay · 250 VAC
24V-FU	+24 V main bus	DN-F10MN	GMC-5	5 A · time-delay · 250 VAC
SR-FU	Safety-relay dedicated feed	DN-F10MN	GMC-3	3 A · time-delay · 250 VAC

Supplementary (GMC) fuses ride behind the current-limiting Class J branch — let-through in a bolted fault is capped upstream.
Blocks used: 6 of 10 DN-F10MN. Elements consumed: 5× GMC-3 (full pack — stock a second pack), 1× GMC-5, 3× JKS-35 (+2 spares).

DWG 01 — POWER ONE-LINE

208 V 3Ø · permanent 4-pole Δ motor on VFD · line reactor deleted (Rev G) · fuse elements per BOM / fuse map (Rev I)

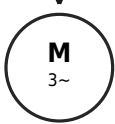
208 V 3PH 60 Hz — wall, 20 A branch → rear line-entry j-box
(front-of-machine AH7810 switch and cable loop DELETED)

MAIN DISCONNECT
Socomec SIRCO M · non-fused rotary · UL 98
wall-mounted S00 handle · padlockable OFF
no door interlock (by design) · line lugs shrouded

BRANCH FUSES — 3× JKS-35
Class J · 35 A · fast-acting · 200 kA IR
in 3× US6J1 single-pole finger-safe blocks

CONTACTOR A — WEG CWBS12
safety contactor · 24 VDC coil
dropped by SAFETY RELAY 13-14 · session chunk

VFD — GS23-22P0
2 HP · 230 V class · 3PH in · CT 7.5 A
R/S/T in · U/V/W out (shielded cable)
e-thermal OL = 6.42 A (replaces heaters)
60 Hz main · 30 Hz preset (LOW) · S-curve
STO ← safety relay 23-24 (jumper removed)
over-torque trip = jam / stall backstop



1.5 HP · 143TC · 6.42 A FLA · PERMANENT 4-pole Δ
(YY 8-pole leads capped in peckerhead) · motor PE + shield to drive-end clamp

PSU-FU — 2× GMC-3
2-pole · 3 A · at tap origin

24 VDC PSU — NDR-240-24
L / N ← both PSU-FU poles (208 V L-L)
+V → GMC-5 (5 A) → +24 bus · -V → 0V

16 AWG tap — fused: 35 A branch can't protect it

KB-FU — 2× GMC-3
2-pole · 3 A · at tap origin

CR-B → CONTACTOR B (theater)
CR-B contact (← Y004 24 VDC coil + diode)
switches vintage Furnas 220 V coil
contacts switch NO LOAD — chunk only

← REV I: both 240 V legs fused (rows 22/22A/24/24A)

MOUNT HIGH in enclosure (Rev I)

DB RESISTOR
BR-N1-280W50 · 50 Ω / 280 W
on + / BR · bleeds reversal energy
thermal sw (N.C.) → PLC X012

PE bonding (all to ground bus)
Enclosure · back panel · door (flex braid) · DISC frame ·
KA + KB frames · DIN rails · VFD PE · motor cable PE +
shield (drive-end clamp only) · MACHINE FRAME / COLUMN
via DEDICATED bond — never via the motor mounting bolts.
Target < 0.1 Ω end-to-end (checklist, sheet 5).

Colors: BLK line/motor (disconnect-controlled) · BLU 24 VDC
control · GN/YE PE. Nothing is live-with-disconnect-OFF
except the DISC line lugs — shrouded (IP20).

Speed reproduction
The VFD synthesizes the original 2:1 pole-change through the
existing belts / Reeves dial: 60 Hz → high ranges (groove A/B);
30 Hz → low range exactly, full V/Hz torque. The operator still
moves the belt and spins the v/s dial; “LOW/HIGH” is the soft
VFD preset (Y003).

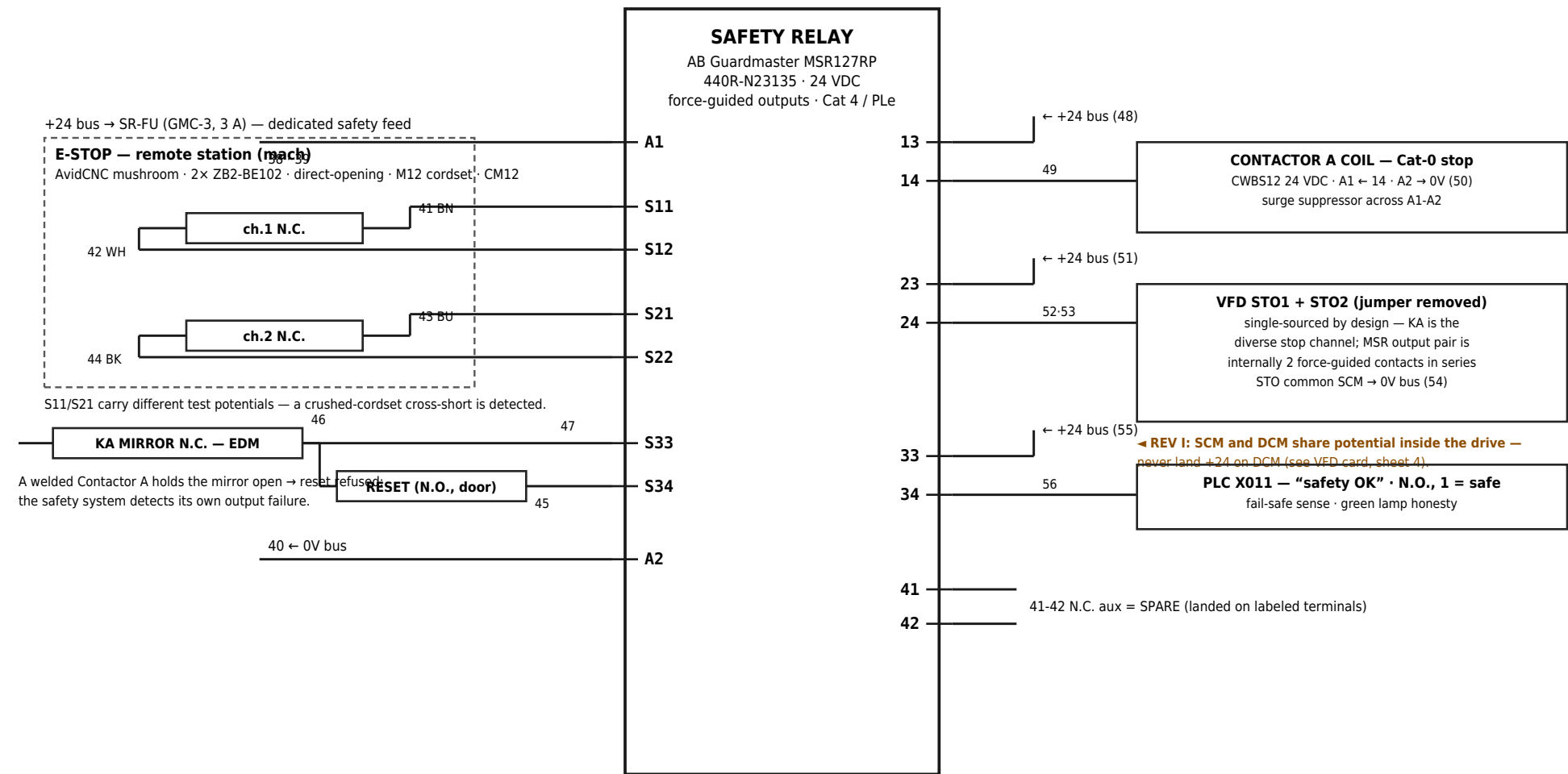
Order matters: LINE → DISC → FUSES → KA → VFD, so locking the
disconnect de-energizes the fuse blocks for fuse changes.

Thermal (Rev I)
NEMA 12 stays SEALED —
no vents (dusty shop).
· DBR mounted high
· internal 24 V stirring
fan (rows 97-98), ~3 W
· commission with a max-
recording thermometer
at the VFD intake.

GoodBetterBestCo — E. Thayer	DWG 01
Power one-line · Powermatic 1200	Rev I · 2026-07-20

DWG 02 — SAFETY RELAY LOOP (firmware-independent)

Dual-channel E-stop · monitored reset (no auto-restart) · EDM detects welded Contactor A · E-stop is the SOLE safety input



Design rules on this drawing

- Safety input is the E-STOP only. No door interlock (wall-mounted handle; door-open work is for qualified persons). No over-travel safety switch: travel is bounded by the depth-rod mechanical stop; a “failed to reverse” fault is a STALL → VFD over-torque trip.
- Two independent stop paths: Contactor A (mechanical power removal) AND VFD STO — plus EDM so a welded A blocks reset.
- Monitored manual RESET → no auto-restart after E-stop or power loss; the PLC additionally requires a fresh FWD press (ladder R13).
- Sequencing limits X006/X007 do NOT enter the safety loop — in TAP they are struck every cycle by design.

GoodBetterBestCo — E. Thayer	DWG 02
Safety relay loop · Powermatic 1200	Rev I · 2026-07-20

VFD PARAMETER CARD — GS23-22P0

Enter with motor uncoupled if practical · parameter numbers per the GS20(X) manual · Rev I supersedes the \$5 table

Function	Setting	Why
Motor rated current	6.42 A (nameplate FLA)	Electronic thermal OL — replaces the heaters
Motor rated freq / voltage	60 Hz / 208 V	V/Hz base point
Command source	External terminals (DI)	PLC brokers all commands
Frequency source	Preset / multi-speed via DI	Y003 selects 30 Hz preset vs 60 Hz main
Main frequency	60 Hz	HIGH range through existing belts
Preset 1 frequency	30 Hz	Reproduces LOW range exactly (full V/Hz torque)
DI1 / DI2 / DI3	Run-FWD / Run-REV / Preset-1	← Y001 / Y002 / Y003 (CLICK sinking outputs)
DI mode ◀ REV I	NPN (SINK), internal power — factory default	See DI wiring box below. Wire 77 is deleted.
Accel / decel + S-curve	Tuned; S-curve ON	Soft start; gentle self-lead tap reversal
Braking (DB)	Chopper enabled; resistor on + / BR	Bleeds reversal energy — no overvolt trip on REV
STO	Wired to safety relay 23-24; factory jumper removed	Fast torque-off; diverse with Contactor A
Over-torque / stall	Enabled; level just above real tapping torque; short delay; action = trip/fault; active 30-60 Hz	THE jam backstop — trips independent of the PLC; verify with a deliberate jam at commissioning

DI wiring — NPN internal power (Rev I; replaces the Rev H “source mode” note)

- NPN/PNP selector: leave at NPN — the factory default. The drive's internal 24 V wets the DI optos.
- The ONLY common conductor is wire 81: CLICK 0 V → DCM. CLICK Y001/Y002/Y003 (sinking) pull DI1/DI2/DI3 to 0 V = active.
- Wire 77 (+24 bus → “DI-COM”) is DELETED. There is no separate DI-common terminal on the GS20 — only DCM.
- NEVER land +24 V on DCM. DCM and the STO common (SCM) share potential inside the drive; with SCM on the 0 V bus (wire 54), external +24 on DCM is a dead short through the drive. This is a documented field failure mode.
- The drive's red “+24V” terminal is exclusively for the factory STO jumper scheme — it stays unused in this design.
- Bench check before first power-up: continuity DCM ↔ CLICK 0 V; NO continuity from any drive terminal to the +24 bus; STO1/STO2 open with E-stop pressed (quick-start procedure).

Sink/source sanity picture

GS20 internal +24 → DI opto → DI1 ← wire 78 → CLICK Y001 (sinks to 0V when ON)
GS20 DCM ← wire 81 → CLICK 0V ↔ 0V bus ↔ SR/STO SCM (wire 54)
(one reference, one common, no +24 anywhere near the drive's control terminals)

COMMISSIONING & VALIDATION — BEFORE POWER-UP, MECHANICAL, VFD

Check off in order · LOTO before any wiring work · safety validation is the part that keeps you alive

Before first power-up

- ☐ Continuity/insulation check all power conductors; no line-level wire reaches any operator control or machine-front lamp.
- ☐ Wiring order: LINE → SIRCO disconnect → 3× US6J1 (JKS-35) → Contactor A → VFD; disconnect line lugs shrouded (live when OFF).
- ☐ Fuse map populated per Shop Pack sheet 1: 3× JKS-35 · 24V-FU GMC-5 · SR-FU GMC-3 · PSU-FU 2× GMC-3 · KB-FU 2× GMC-3.
- ☐ Contactor-B tap fused at origin: FB L1 → KB-FU → CR-B COM, and FB L2 → KB-FU → KB A2 (wire rows 22/22A/24/24A).
- ☐ PE bonding < 0.1 Ω end-to-end incl. DEDICATED machine frame/column bond (not motor bolts), rails, door braid, both contactors.
- ☐ 24 V field polarity per I/O table; prox (ICF12, PNP-NO) on the sinking common: BN→+24 (73) · BU→0V (74) · BK→X010 (75).
- ☐ GS20 DI check: NPN switch at factory default; DCM ↔ CLICK 0V continuity (81); wire 77 ABSENT; no drive terminal to +24 bus.
- ☐ DB resistor mounted high; stirring fan wired (97/98) and spinning whenever the 24 V bus is up; enclosure fully sealed.

Mechanical

- ☐ Depth-rod METAL end stop takes the bottom load — rod lands on the stop, not a limit-switch lever, both directions.
- ☐ Set the stop first; then bottom/top limit tags slightly shy of it.
- ☐ Prox target verified ferrous/clean at the feed-lever OFF detent; X010 = 1 with lever OFF, 0 with lever engaged or prox unplugged.

VFD setup (motor uncoupled if practical) — use the VFD parameter card, sheet 4

- ☐ FLA 6.42 A entered; electronic thermal OL enabled.
- ☐ 60 Hz main / 30 Hz preset-1; Y003 toggles the preset.
- ☐ Accel/decel + S-curve; DB chopper on; thermal switch to X012 proven: force it open → blink-red, self-clears after cool-down.
- ☐ Over-torque trip proven with a deliberate jam (clamped stock) — drive faults without PLC involvement.
- ☐ Jog FWD → verify spindle direction; if reversed swap two VFD OUTPUT leads, never input.
- ☐ Power-cycle etiquette: use E-stop/STOP for stops — don't cycle the disconnect repeatedly (DC-bus precharge wear).

GoodBetterBestCo — E. Thayer	CHECKLIST
Commissioning · Powermatic 1200	Rev I · 2026-07-20

COMMISSIONING & VALIDATION — SAFETY, INDICATORS, SEQUENCES

Check off in order · LOTO before any wiring work · safety validation is the part that keeps you alive

Contactors & safety validation

- ☐ Both overload heaters removed; motor phases straight through. Contactor B contacts switch NO load; CR-B diode fitted.
- ☐ E-stop → Contactor A drops (audible chunk) AND STO opens; motor stops. Reset required; power-cycle comes up stopped.
- ☐ EDM: jumper Contactor A closed (simulated weld) → safety relay refuses reset. Remove jumper, reset accepted.
- ☐ X011 tracks safety state at the PLC; green drops the instant safety trips.

Indicators (four states, exactly one active)

- ☐ Safe + no fault + not jogging → green solid. Arm jog → green blink. Open X012 → red blink, self-clears. →X011 or C10 → red solid.
- ☐ Over-temp AND hard fault together → red solid wins. Never two lamps, never neither, while powered. White lamp = bus, PLC-independent.

DRILL (bidirectional, stop-first)

- ☐ FWD press → latched run at drum speed; STOP ramps down. REV press → latched reverse; same one-press model.
- ☐ Running FWD, one REV press STOPS (no reversal); second REV press runs reverse. Mirror-check FWD-stops-REV.
- ☐ Rapid opposite taps never plug-reverse; no stray run from the FWD release after an opposite-stop (C50 path).
- ☐ Bottom limit while running FWD → fault + red solid. Reverse into the top stop → VFD over-torque trips (not a ladder fault).

TAP (single-tap)

- ☐ Entry gate: lever engaged → FWD does not start. Lever OFF: FWD → down → bottom limit → ramped reverse → top limit → stop.
- ☐ Bump lever engaged mid-cycle → immediate fault-stop.
- ☐ E-stop mid-cycle, then reset → stays in IDLE; a fresh FWD press is required (R13 →C20 abort).

JOG mode (Rev I behavior)

- ☐ **With safety tripped and drum at LOW/HIGH, C20 stays OFF (verifies the R1 series fix).**
- ☐ **Hold FWD ≥5 s, drum ON: green blinks at 5 s and the spindle does NOT turn while the entry press is held;**
- ☐ **release, then a fresh FWD press jogs at 30 Hz (verifies →C50 in R24).**
- ☐ **Drum OFF: hold FWD ≥5 s → jog arms; presses do nothing; drum to LOW → presses jog; drum OFF mid-hold → stops (C61).**
- ☐ **Short STOP press exits jog; long STOP hold stops motion but stays armed; 60 s idle disarms; any fault or E-stop force-exits.**
- ☐ While armed, normal latched runs and TAP starts are inhibited.

Chunk & thermal

- ☐ Contactor B fires ~300 ms on each run start / direction change (DRILL and TAP); Contactor A does NOT cycle per-operation.
- ☐ Max-recording thermometer at the VFD intake through a real tapping session; escalate cooling only if it passes ~50 °C.

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